1(a). $p = 7$

$q = 11$

(b) $n = pq = 77$

public key

$(e, n) = (13, 77)$

(c) $(p-1)(q-1) = 60$

$\gcd(13, 60) = 1$

$e = 13$

(d) $13d \equiv 1 \pmod{60}$

$\gcd(13, 60) = 1 \Rightarrow 13s + 60t = 1$

$13s + 0 = 1 \pmod{60}$

$13s = 1 \pmod{60}$

now $s = d$ which we are looking for

we can use pulverizer to compute $\gcd(13, 60) \Rightarrow 13s + 60t = 1$

By Euler's theorem, $\gcd(n, m) = 1$,

$$m^{\phi(n)} = 1 \pmod{n}$$

$\gcd(13, 60) = 1$

$$13^{\phi(60)} = 1 \pmod{60}$$

$$13^{16} = 1 \pmod{60}$$

$$13^{15} \cdot 13 = 1 \pmod{60}$$

$$13^{15} \cdot e = 1 \pmod{60}$$

$d = 13^{15} \pmod{60}$

$13^{15} = 13^8 \cdot 13^4 \cdot 13^2 \cdot 13$

$= 1 \cdot 1 \cdot 49 \cdot 13$

$= 637 \pmod{60}$

$= 37 \pmod{60}$

$d = 37$

$60 = 3 \times 5 \times 2^2$

$\phi(60) = 60 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{5}) \times (1 - \frac{1}{3})$

$= 60 \times \frac{1}{2} \times \frac{4}{5} \times \frac{2}{3}$

$= 16$

power	result	mod 60
13^2	169	49
13^4	$(49)^2 = 2401$	1
13^8	$(1)^2$	1

(2) $m = 2$

$m' = \text{rem}(2^{13}, 77)$

$= 30$

$m = \text{rem}(30^{37}, 77)$

$= 2$

$2^2 \quad 4 \quad 4$

$2^4 \quad 4^2 \quad 16$

$2^8 \quad 16^2 \quad 25$

$2^{13} = 2^8 \cdot 2^4 \cdot 2$

$= 25 \cdot 16 \cdot 4$

$= 800$

$= 30 \pmod{77}$

⑤ if you fact n int p and q you get d by computing

$$de \equiv 1 \pmod{(p-1)(q-1)}$$